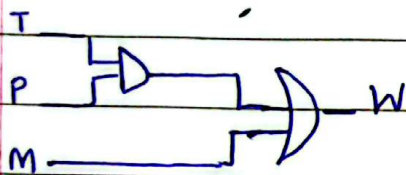


Assignment - 03

Q.1

M	T	P	W	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$M' T P$
1	0	0	1	$M T' P'$
1	0	1	1	$M T' P$
1	1	0	1	$M T P'$
1	1	1	1	$M T P$

Boolean: $W = M + (\bar{M} \cdot T \cdot P) = M + (T \cdot P)$



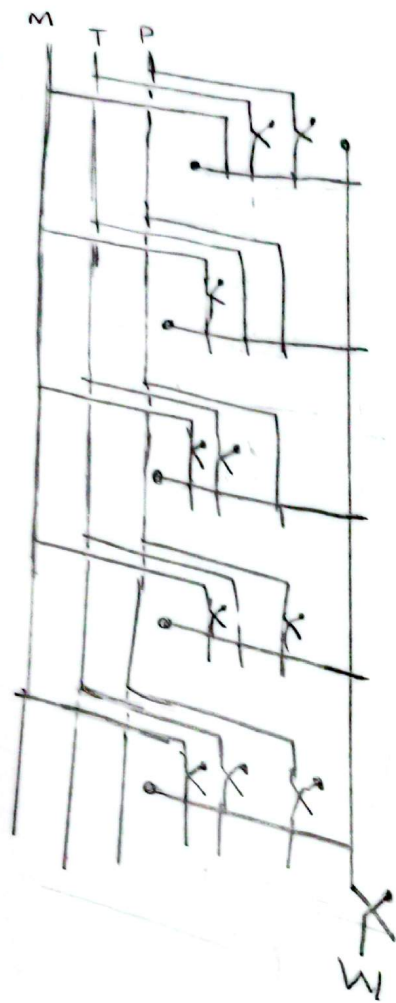
Q.2

W	R	B	P	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$W' R B$
1	0	0	1	$W R' B'$
1	0	1	1	$W R' B$
1	1	0	1	$W R B'$
1	1	1	1	$W R B$

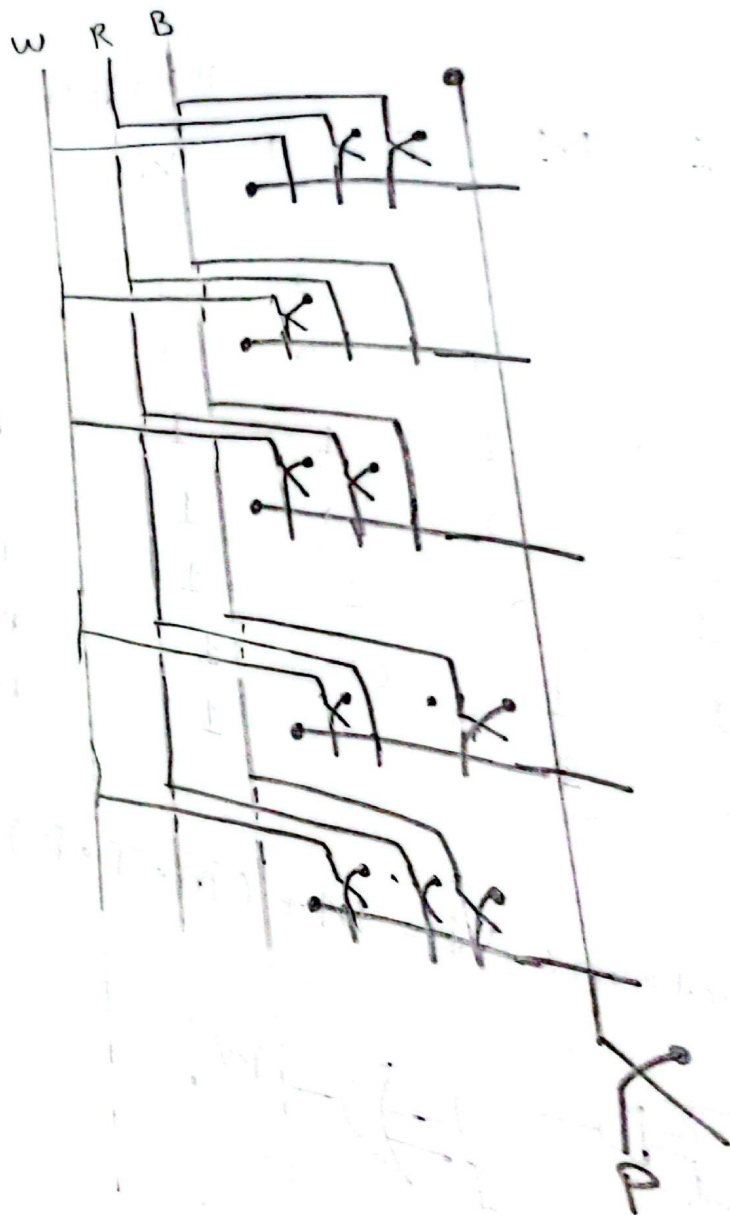
Boolean: $P = W + R \cdot B$

2.9. m r

1c.

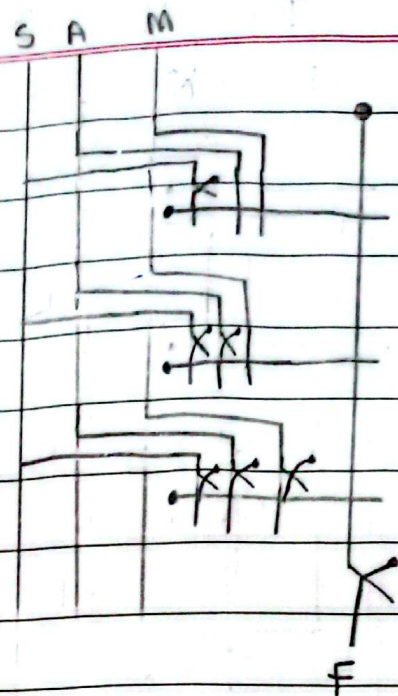


2.C



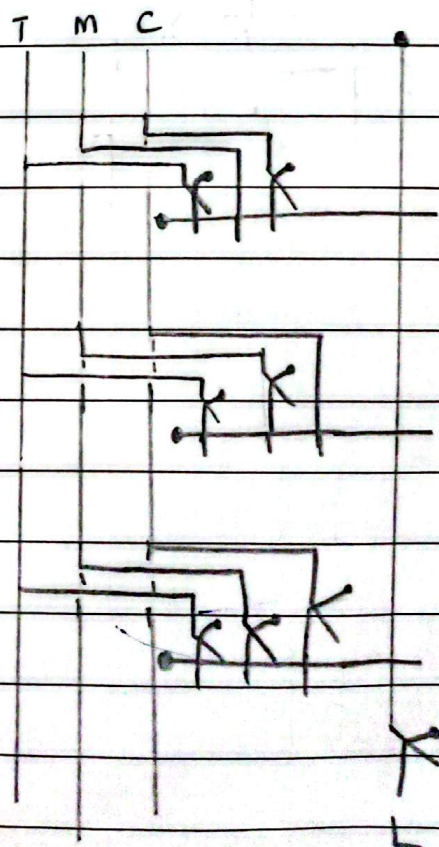
Q.3.

S	A	M	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

(1) $SA'M'$ (1) SAM' (1) SAM Boolean: $F = S\bar{M} + SA = S(\bar{M} + A)$

Q.4.

T	M	C	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

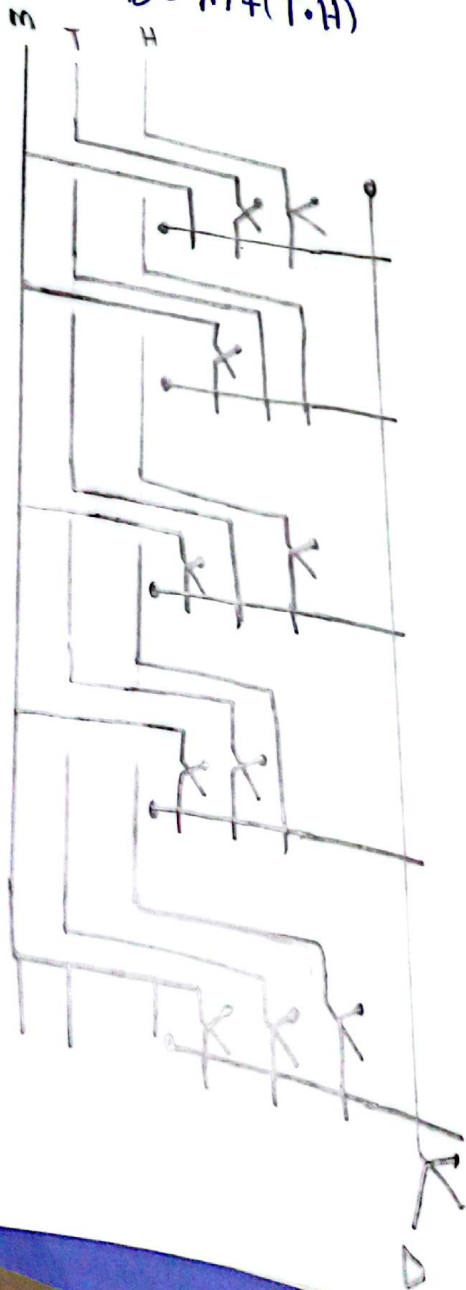
 $T\bar{M}C$ $TM\bar{C}$ TMC Boolean: $L = T \cdot (M + C)$

Q.5.

M	T	H	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$M' T H$
 $M T' H'$
 $M T' H$
 $M T H'$
 $M T H$

Boolean: $D = M + (T \cdot H)$



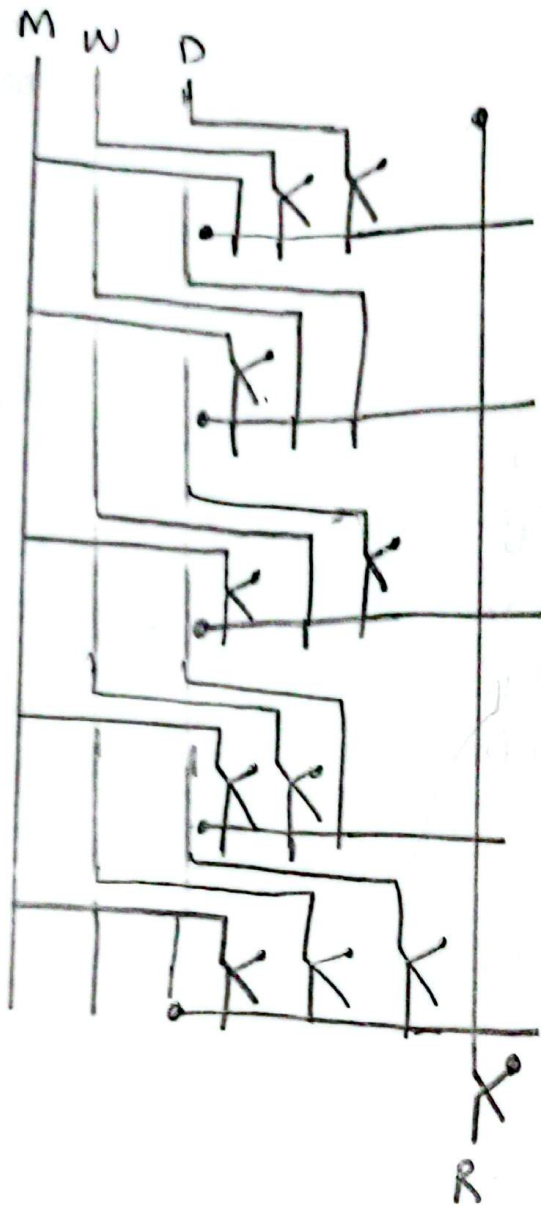
Q.6.	M	W	D	R	
	0	0	0	0	
	0	0	1	0	
	0	0	0	0	
	0	1	1	1	$M'WD$
	1	0	0	1	$MW'D'$
	1	0	1	1	$MW'D$
	1	1	0	1	MWD'
	1	1	1	1	MWD

Boolean: $R = M + WD$

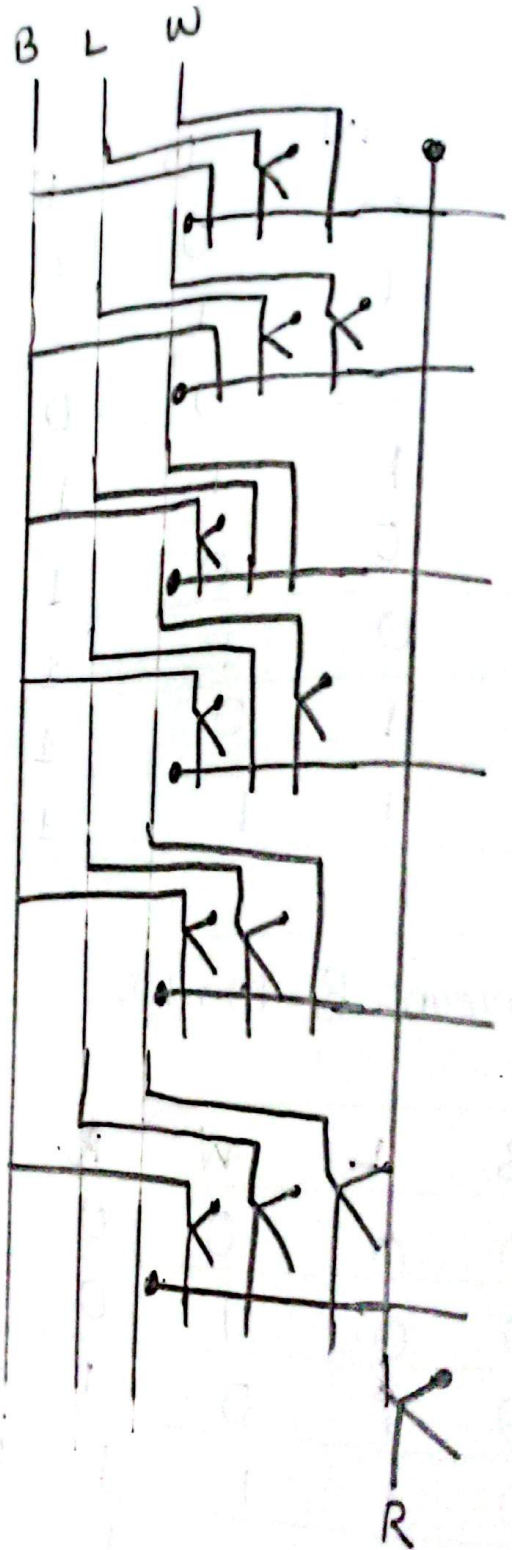
Q.7	B	L	W	R	
	0	0	0	0	
	0	0	1	0	
	0	1	0	1	$B'LW'$
	0	1	1	1	$B'LW$
	1	0	0	1	$BL'W'$
	1	0	1	1	$BL'W$
	1	1	0	1	BLW'
	1	1	1	1	BLW

Boolean: $R = B + L + (W \cdot L)$
 $= B + L$

6.C.



7.C



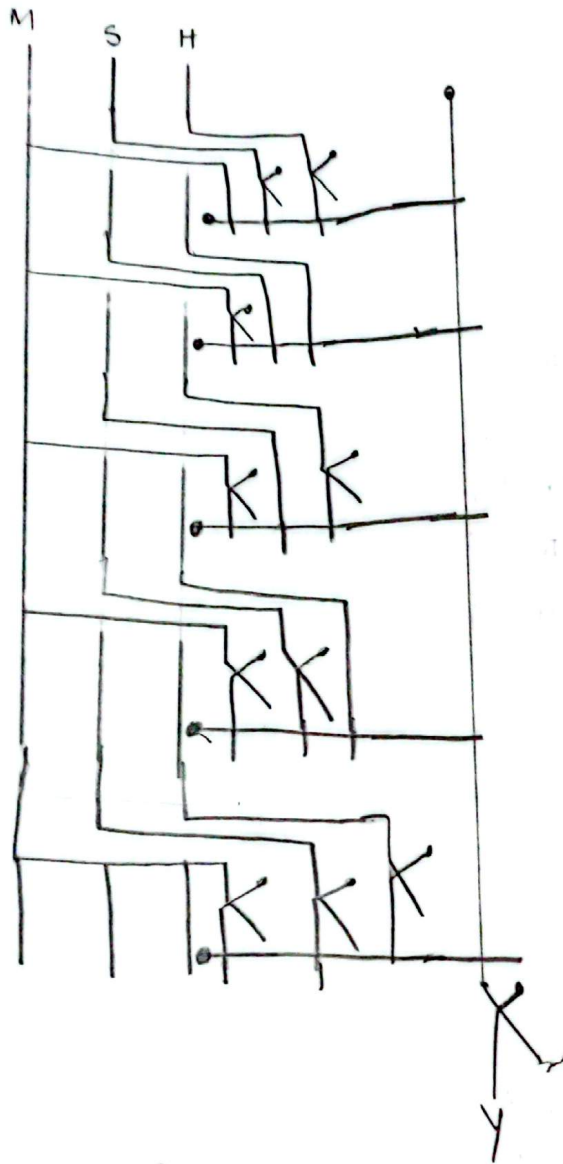
Q.8	M	S	H	Y	
	0	0	0	0	
	0	0	1	0	
	0	1	0	0	
	0	1	1	1	$M'SH$
	1	0	0	1	$MS'H'$
	1	0	1	1	$MS'H$
	1	1	0	1	MSH'
	1	1	1	1	MSH

Boolean: $Y = M + SH$

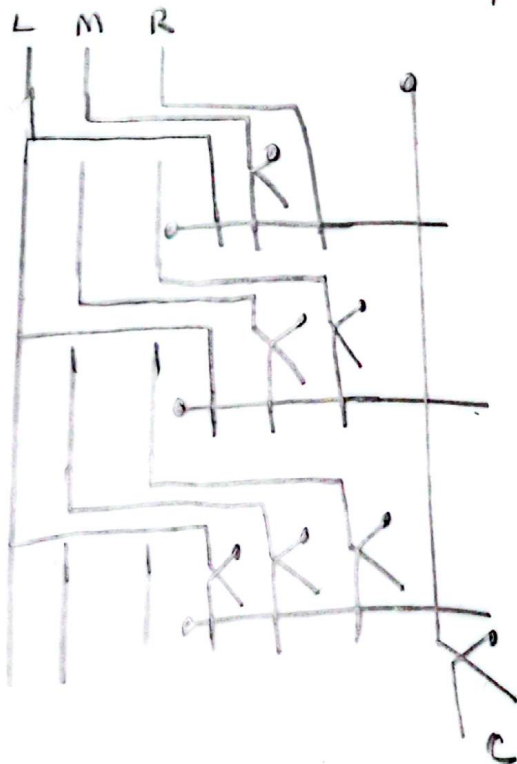
Q.9.	L	M	R	C	
	0	0	0	0	
	0	0	1	0	
	0	1	0	1	$L'MR'$
	0	1	1	1	$L'MR$
	1	0	0	0	
	1	0	1	0	
	1	1	0	0	
	1	1	1	1	LMR

$$\begin{aligned}
 \text{Boolean: } C &= \bar{L}M + LRM \\
 &= M(\bar{L} + LR) \\
 &= M(\bar{L} + R) \\
 &= \bar{L}M + MR
 \end{aligned}$$

8.c.



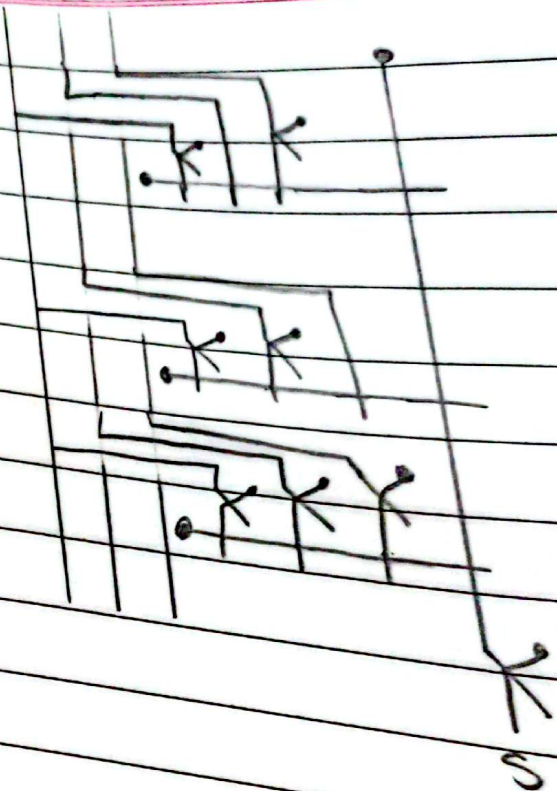
9.c.



Q.10.

B	D	W	S	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	0	
1	0	0	0	
1	0	1	0	
1	1	0	1	$BD'W$
1	1	1	1	BDW'
			1	BDW

B D W



$$\text{Boolean: } BD + BW = S$$

$$S = B(D + W)$$